

Chapter 14: High-Yield NBDHE Study Guide

Periodontics & Clinical Periodontology (English)

Periodontics & Clinical Periodontology

1. Periodontal Anatomy & Biologic Width

- **Attached Gingiva:**
 - Width: Widest in the **Maxillary Anterior** (3.5–4.5 mm); narrowest in the **Mandibular Premolar Facial** (1.8 mm).
 - Boundary: Separated from alveolar mucosa by the **Mucogingival Junction (MGJ)** (present on facial and mandibular lingual, absent on palate).
- **Junctional Epithelium (JE):**
 - Length: **0.25 to 1.35 mm** (average $\approx$ 0.97 mm).
 - Attachment: Attached to tooth via **hemidesmosomes** and the internal basal lamina. Highly permeable, providing passage for GCF and PMNs.
- **Alveolar Bone Crest:**
 - Normal position: Located **1.5 to 2.0 mm apical** to the cemento-enamel junction (CEJ) in health. Follows the scalloped contour of the CEJ.
- **Bone Defects:**
 - **Fenestration:** Isolated "window" of bone loss leaving root surface exposed, with marginal bone intact.
 - **Dehiscence:** Cleft-like bone defect extending continuously from the marginal bone crest apically.

2. Etiology & Pathogenesis of Periodontal Diseases

- **Microbiology & Subgingival Biofilm:**
 - Earliest immune responders in Stage I (Initial) Gingivitis: **Neutrophils (PMNs)** via chemotaxis.
 - Apical subgingival attached biofilm: Dominated by anaerobic **Gram-negative rods and filaments** (e.g., *Porphyromonas gingivalis*, *Tannerella forsythia*, *Treponema denticola* — Red Complex).
 - Calculus: Acts as a **plaque-retentive factor**, not the primary chemical initiator.
- **Medication-Induced Gingival Enlargement:**
 - Calcium Channel Blockers (e.g., **Nifedipine**, Amlodipine).
 - Immunosuppressants (e.g., **Cyclosporine**).
 - Anticonvulsants (e.g., **Phenytoin / Dilantin**).
 - Most frequently affects the **anterior labial/interproximal** gingiva.
- **Necrotizing Periodontal Diseases (NPD):**
 - Punched-out crater-like interdental papillae, pseudomembranous slough, fetid odor, pain, and bleeding.

3. Nonsurgical Periodontal Therapy (NSPT) & Re-evaluation

- **Objective of NSPT:** To eliminate etiologic biofilm/calculus, reduce inflammation, and facilitate biological healing.
- **Optimal Treatment Response:** Initial probing depths of **4 to 6 mm** achieve the highest rate of probing depth reduction and clinical attachment gain.

- **Re-evaluation Timing:** **4 to 6 weeks** post-instrumentation is required to allow connective tissue reorganization, epithelial junctional adaptation, and resolution of inflammation.
- **Gingival Condition / Bleeding on Probing (BOP):** The best clinical indicator of periodontal stability or active disease progression.

4. Periodontal Surgery & Dental Implants

- **Guided Tissue Regeneration (GTR):** Placement of a barrier membrane to prevent epithelial downgrowth, allowing slower-migrating PDL and osteogenic cells to populate the root surface.
- **Suture Removal:** Grasp the knot with cotton pliers, gently lift, and cut close to the tissue below the knot to prevent pulling contaminated external suture material through the wound.
- **Implant Biology & Maintenance:**
 - **Perimucosal Seal:** Epithelial adaptation forming a biological seal around the titanium abutment cylinder.
 - Normal 1st-Year Bone Loss: **1 to 2 mm** during healing/remodeling, followed by <0.2 mm annually.
 - 5-Year Success Rate: $\geq 85\%$.
 - Cleaning: Use non-scratching plastic, resin, graphite, or titanium instruments.
 - **Antimicrobial Fluoride:** **Stannous Fluoride (SnF_2)** possesses direct antimicrobial action against plaque pathogens via the stannous ion.